

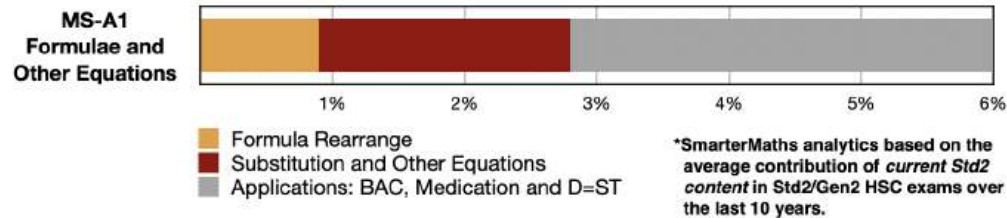
STD 1: Algebra (Std 1), A1 Formulae and Equations (Y11)

Applications: BAC, Medication and D=SxT (Std 1)

Teacher: Denis Vlismas

Exam Equivalent Time: 54 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Gen2/Std2 Exam Contribution History A1 Formulae and Equations



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- MS-A1 Formulae and Equations has accounted for 6% and 5% of the 2020 and 2019 Std1 exams respectively.
- MS-A1 Formulae and Equation is split into the sub-topics listed in the graph above. Note that estimated exam contributions in the graph are taken from Std2/Gen2 past exams due to the very limited Std1 exam history. They should only be used for estimating the "relative" importance of each sub-topic.
- This analysis looks at Applications: BAC, Medication and D=SxT.

ANALYSIS - What to Expect and Common pitfalls

- MS-A1 Formulae and Other Equations has been examined with just a single mark multiple choice question in both the 2020 and 2019 Std1 exams.
- Although BAC calculations have been asked in 6 of the last 7 Std2/Gen2 exams, it has yet to appear in the Std1 exam. An important revision area.
- Substitution into Fried's, Young's and Clark's dosage formulae is also consistently tested in the Gen2/Std2 exams but not in the Std1 paper to date.
- Stopping distance is another prime candidate question type that hasn't appeared in the first two Std1 exams (noting that 2020 Std1 18 was a formula rearrange and substitution question). Again, we recommend a revision focus here.
- Distance, speed and time equations ($D=S \times T$) have been the basis of challenging multiple choice questions in the past. We include numerous past questions in the database, acknowledging the obvious overlap with M4 Rates within the measurement strand. 2011 Std2 21 MC is a great revision example of the higher difficulty level in this area.

Questions

1. Algebra, STD2 A1 2017 HSC 2 MC

A car is travelling at 95 km/h.

How far will it travel in 2 hours and 30 minutes?

- A. 38 km
- B. 41.3 km
- C. 218.5 km
- D. 237.5 km

2. Algebra, STD2 A1 SM-Bank 1 MC

The blood alcohol content (**BAC**) of a male's blood is given by the formula;

$$BAC_{\text{male}} = \frac{10N - 7.5H}{6.8M}, \text{ where}$$

N is the number of standard drinks consumed,

H is the number of hours drinking and

M is the person's mass in kgs.

Calculate the **BAC** of a male who consumed 4 standard drinks in 3.5 hours and weighs 68 kgs, correct to 2 decimal places.

- (A) 1.03
- (B) 0.03
- (C) 0.04
- (D) 0.01

3. Algebra, STD2 A1 2012 HSC 15 MC

A car takes 6 hours to complete a journey when travelling at 60 km/h.

How long would the same journey take if the car were travelling at 100 km/h?

- (A) 36 minutes
- (B) 1 hour and 40 minutes
- (C) 3 hours and 6 minutes
- (D) 3 hours and 36 minutes

4. Algebra, STD1 A1 2020 HSC 7 MC

The distance between Bricktown and Koala Creek is 75 km. A person travels from Bricktown to Koala Creek at an average speed of 50 km/h.

How long does it take the person to complete the journey?

- A. 40 minutes
- B. 1 hour 25 minutes
- C. 1 hour 30 minutes
- D. 1 hour 50 minutes

5. Algebra, STD2 A1 2015 HSC 23 MC

The number of ‘standard drinks’ in various glasses of wine is shown.

Number of standard drinks			
White Wine		Red Wine	
small glass	large glass	small glass	large glass
0.9	1.4	1.0	1.5

A woman weighing 62 kg drinks three small glasses of white wine and two large glasses of red wine between 8 pm and 1 am.

Using the formula for calculating blood alcohol below, what would be her blood alcohol content (BAC) estimate at 1 am, correct to three decimal places?

$$BAC_{\text{Female}} = \frac{10N - 7.5H}{5.5M}$$

where *N* is the number of standard drinks consumed
H is the number of hours drinking
M is the person's mass in kilograms

- (A) 0.030
- (B) 0.037
- (C) 0.046
- (D) 0.057

6. Algebra, STD2 A1 2016 HSC 10 MC

Caroline drinks two small bottles of wine over a three-hour period. Each of these bottles contains 2.3 standard drinks. Caroline weighs 53 kg.

Using the formula below, what is Caroline's approximate blood alcohol content (BAC) at the end of this period?

$$BAC_{\text{Female}} = \frac{10N - 7.5H}{5.5M}$$

where *N* is the number of standard drinks consumed
H is the number of hours drinking
M is the person's mass in kilograms

- (A) 0.081
- (B) 0.065
- (C) 0.0017
- (D) 0.0014

7. Algebra, STD2 A1 2017 HSC 19 MC

Young's formula, shown below, is used to calculate the dosage of medication for children aged 1–12 years based on the adult dosage.

$$D = \frac{yA}{y + 12}$$

where *D* = dosage for children aged 1–12 years
y = age of child (in years)
A = Adult dosage

A child's dosage is calculated to be 20 mg, based on an adult dosage of 40 mg.

How old is the child in years?

- A. 6
- B. 8
- C. 10
- D. 12

8. Algebra, STD2 A1 2011 HSC 21 MC

A train departs from Town A at 3.00 pm to travel to Town B. Its average speed for the journey is 90 km/h, and it arrives at 5.00 pm. A second train departs from Town A at 3.10 pm and arrives at Town B at 4.30 pm.

What is the average speed of the second train?

- (A) 135 km/h
- (B) 150 km/h
- (C) 216 km/h
- (D) 240 km/h

9. Algebra, STD1 A1 2019 HSC 8 MC

Heart rate is measured in beats per minute. Maximum heart rate (MHR) is calculated using the following formula.

$$\text{MHR} = 220 - \text{age}$$

Target heart rates are calculated as a percentage of MHR.

Felicity's age is 28. Her trainer calculates that her target heart rate range is 60% to 80% of her MHR.

Which of the following lies within this target heart rate range?

- A. 100 beats per minute
- B. 140 beats per minute
- C. 180 beats per minute
- D. 220 beats per minute

10. Algebra, STD2 A1 2014 HSC 4 MC

Young's formula below is used to calculate the required dosages of medicine for children aged 1–12 years.

$$\text{Dosage} = \frac{\text{age of child (in years)} \times \text{adult dosage}}{\text{age of child (in years)} + 12}$$

How much of the medicine should be given to an 18-month-old child in a 24-hour period if each adult dosage is 45 mL? The medicine is to be taken every 6 hours by both adults and children.

- (A) 5 mL
- (B) 20 mL
- (C) 27 mL
- (D) 30 mL

11. Algebra, STD2 A1 2018 HSC 26b

Clark's formula, given below, is used to determine the dosage of medicine for children.

$$\text{Dosage} = \frac{\text{weight in kg} \times \text{adult dosage}}{70}$$

For a particular medicine, the adult dosage is 325 mg and the correct dosage for a specific child is 90 mg.

How much does the child weigh, to the nearest kg? (2 marks)

12. Algebra, STD2 A1 SM-Bank 5

Fried's formula is used to calculate the medicine dosages for children aged 1-2 years.

$$\text{Child dosage} = \frac{\text{Age(in months)} \times \text{adult dosage}}{150}$$

Ben is 1.5 years old and receives a daily dosage of 450 mg of a medicine.

According to Fried's formula, what would the appropriate adult daily dosage of the medicine be? (2 marks)

13. Algebra, STD2 A1 2015 HSC 26b

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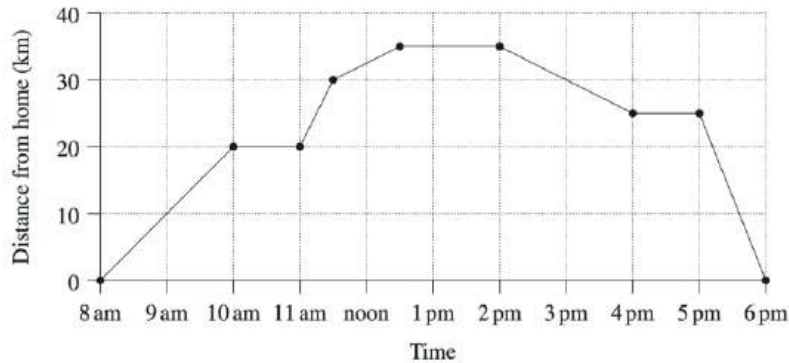
The adult daily dosage of a medicine contains 3150 mg of a particular drug.

A child who weighs 35 kg is to be given tablets each containing 525 mg of this drug.

How many tablets should this child be given daily? (2 marks)

14. Algebra, STD2 A4 2013 HSC 27a

Lucy went for a bike ride. She left home at 8 am and arrived back at home at 6 pm. A graph representing her journey is shown.



- What was the total distance that she rode during the day? *(1 mark)*
- How much time did Lucy spend riding her bike during the day? *(1 mark)*

15. Algebra, STD2 A1 2014 HSC 29b

Blood alcohol content of males can be calculated using the following formula

$$BAC_{\text{Male}} = \frac{10N - 7.5H}{6.8M}$$

where N is the number of standard drinks consumed

H is the number of hours drinking

M is the person's mass in kilograms

What is the maximum number of standard drinks that a male weighing 84 kg can consume over 4 hours in order to maintain a blood alcohol content (BAC) of less than 0.05? *(3 marks)*

16. Algebra, STD2 A1 2017 HSC 27e

Rhys is drinking low alcohol beer at a party over a five-hour period. He reads on the label of the low alcohol beer bottle that it is equivalent to 0.8 of a standard drink.

Rhys weighs 90 kg.

The formula below can be used to calculate a Rhys's blood alcohol content.

$$BAC_{\text{Male}} = \frac{10N - 7.5H}{6.8M}$$

where N is the number of standard drinks consumed

H is the number of hours drinking

M is the person's mass in kilograms

What is the maximum number of complete bottles of the low alcohol beer he can drink to remain under a Blood Alcohol Content (BAC) of 0.05? *(4 marks)*

17. Algebra, STD2 A1 2005 HSC 24b

The formula $D = \frac{2A}{15}$ is used to calculate the dosage of Hackalot cough medicine to be given to a child.

- D is the dosage of Hackalot cough medicine in millilitres (mL).
- A is the age of the child in months.

- If George is nine months old, what dosage of Hackalot cough medicine should he be given? *(1 mark)*

The correct dosage of Hackalot cough medicine for Sam is 4 mL.

- What is the difference in the ages of Sam and George, in months? *(3 marks)*

18. Algebra, STD2 A1 2018 HSC 28e

Sophie is driving at 70 km/h. She notices a branch on the road ahead and decides to apply the brakes. Her reaction time is 1.5 seconds. Her braking distance (D metres) is given by $D = 0.01v^2$, where v is speed in km/h.

Stopping distance can be calculated using the following formula

$$\text{stopping distance} = \{\text{reaction time distance}\} + \{\text{braking distance}\}$$

What is Sophie's stopping distance, to the nearest metre? *(3 marks)*

19. Algebra, STD2 A1 SM-Bank 4

Doris is driving in a school zone at a speed of 35 kilometres per hour and needs to stop immediately to avoid an accident.

It takes her 1.25 seconds to react and her braking distance is 5.3 metres.

Stopping distance can be calculated using the following formula

$$\text{stopping distance} = \{\text{reaction time distance}\} + \{\text{braking distance}\}$$

What is Doris' total stopping distance? Give your answer to 1 decimal place. (2 marks)

20. Algebra, STD2 A1 2015 HSC 30d

Claire is driving on a motorway at a speed of 110 kilometres per hour and has to brake suddenly. She has a reaction time of 2 seconds and a braking distance of 59.2 metres.

Stopping distance can be calculated using the following formula

$$\text{stopping distance} = \{\text{reaction time distance}\} + \{\text{braking distance}\}$$

What is Claire's stopping distance. (2 marks)

Worked Solutions

1. Algebra, STD2 A1 2017 HSC 2 MC

$$\begin{aligned}\text{Distance} &= 95 \times 2.5 \\ &= 237.5 \text{ km} \\ \Rightarrow D\end{aligned}$$

2. Algebra, STD2 A1 SM-Bank 1 MC

$$\begin{aligned}BAC_{\text{male}} &= \frac{(10 \times 4) - (7.5 \times 3.5)}{(6.8 \times 68)} \\ &= \frac{13.75}{462.4} \\ &= 0.0297... \\ \Rightarrow B\end{aligned}$$

3. Algebra, STD2 A1 2012 HSC 15 MC

$$\begin{aligned}T &= \frac{D}{S} \\ \text{Since } T &= 6 \text{ when } S = 60 \\ 6 &= \frac{D}{60} \\ D &= 360 \text{ km}\end{aligned}$$

Find T when $S = 100$ and $D = 360$

$$\begin{aligned}T &= \frac{360}{100} \\ &= 3.6 \text{ hours} \\ &= 3 \text{ hrs } 36 \text{ minutes} \\ \Rightarrow D\end{aligned}$$

4. Algebra, STD1 A1 2020 HSC 7 MC

$$\begin{aligned}\text{Time} &= \frac{\text{Distance}}{\text{Speed}} \\ &= \frac{75}{50} \\ &= 1.5 \text{ hours} \\ &= 1 \text{ hour } 30 \text{ minutes}\end{aligned}$$

$\Rightarrow C$

Mean mark 52%.

5. Algebra, STD2 A1 2015 HSC 23 MC

$$\begin{aligned}N &= 3 \times 0.9 + 2 \times 1.5 \\ &= 5.7 \text{ standard drinks} \\ H &= 5 \text{ hours} \\ M &= 62 \text{ kg}\end{aligned}$$

$$\begin{aligned}\therefore BAC_f &= \frac{10 \times 5.7 - 7.5 \times 5}{5.5 \times 62} \\ &= 0.05718 \dots\end{aligned}$$

$\Rightarrow D$

6. Algebra, STD2 A1 2016 HSC 10 MC

$$\begin{aligned}BAC_f &= \frac{10N - 7.5H}{5.5M} \\ &= \frac{10(2 \times 2.3) - 7.5(3)}{5.5 \times 53} \\ &= 0.0806 \dots\end{aligned}$$

$\Rightarrow A$

7. Algebra, STD2 A1 2017 HSC 19 MC

$$D = \frac{yA}{y + 12}$$

$$20 = \frac{40y}{y + 12}$$

$$20(y + 12) = 40y$$

$$20y + 240 = 40y$$

$$20y = 240$$

$$y = 12$$

$\Rightarrow D$

8. Algebra, STD2 A1 2011 HSC 21 MC

1st train

Travels 2hrs at 90km/h

Distance = Speed $\times$ Time

$$= 90 \times 2$$

$$= 180 \text{ km}$$

2nd train

Travels 180 km in 1 hr 20 min $\left(\frac{4}{3} \text{ hrs}\right)$

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$= 180 \div \frac{4}{3}$$

$$= 180 \times \frac{3}{4}$$

$$= 135 \text{ km/h}$$

$\Rightarrow A$

♦ Mean mark 49%

9. Algebra, STD1 A1 2019 HSC 8 MC

$$\begin{aligned}\text{MHR} &= 220 - 28 \\ &= 192 \\ 60\% \times 192 &= 115.2 \\ 80\% \times 192 &= 153.6\end{aligned}$$

140 beats is in the range.
i.e. $115.2 < 140 < 153.6$
 $\Rightarrow B$

♦ Mean mark 46%.

10. Algebra, STD2 A1 2014 HSC 4 MC

$$\begin{aligned}\text{Dosage} &= \frac{1.5 \times 45}{1.5 + 12} \\ &= 5 \text{ mL}\end{aligned}$$

♦ Mean mark 42%

Since 1 dosage every 6 hrs
In 24 hours,
Medicine given $= 4 \times 5 = 20 \text{ mL}$
 $\Rightarrow B$

11. Algebra, STD2 A1 2018 HSC 26b

$$\begin{aligned}90 &= \frac{\text{weight} \times 325}{70} \\ \therefore \text{weight} &= \frac{70 \times 90}{325} \\ &= 19.38\dots \\ &= 19 \text{ kg (nearest kg)}\end{aligned}$$

12. Algebra, STD2 A1 SM-Bank 5

Substituting into the formula:

$$\begin{aligned}450 &= \frac{18 \times \text{adult dosage}}{150} \\ \therefore \text{Adult dosage} &= \frac{450 \times 150}{18} \\ &= 3750 \text{ mg}\end{aligned}$$

13. Algebra, STD2 A1 2015 HSC 26b

$$\begin{aligned}\text{Dosage} &= \frac{35 \times 3150}{70} \\ &= 1575 \text{ mg}\end{aligned}$$

$$\begin{aligned}\# \text{ Tablets per day} &= \frac{\text{Dosage}}{\text{mg per tablet}} \\ &= \frac{1575}{525} \\ &= 3\end{aligned}$$

$\therefore$ The child should be given 3 tablets per day.

14. Algebra, STD2 A4 2013 HSC 27a

$$\begin{aligned}\text{i. Total distance} &= 35 + 35 \\ &= 70 \text{ km}\end{aligned}$$

$$\begin{aligned}\text{ii. Time away from home} &= 10 \text{ hours} \\ \text{Time resting} &= 1 + 1.5 + 1 \\ &= 3.5 \text{ hours}\end{aligned}$$

$$\begin{aligned}\therefore \text{Time riding} &= 10 - 3.5 \\ &= 6.5 \text{ hours}\end{aligned}$$

15. Algebra, STD2 A1 2014 HSC 29b

$$\text{BAC}_{\text{male}} = \frac{10N - 7.5H}{6.8M}$$

Find N for $\text{BAC} < 0.05$, given $H = 4$ and $M = 84$

$$\frac{10N - 7.5(4)}{6.8(84)} < 0.05$$

$$10N - 30 < 0.05(571.2)$$

$$10N < 28.56 + 30$$

$$< 58.56$$

$$N < 5.856$$

$\therefore$ Max number of drinks is 5.

16. Algebra, STD2 A1 2017 HSC 27e

$$\text{BAC}_{\text{male}} = \frac{10N - 7.5H}{6.8M}$$

$$0.05 = \frac{10N - 7.5 \times 5}{6.8 \times 90}$$

$$10N = (0.05 \times 6.8 \times 90) + 7.5 \times 5$$

$$= 68.1$$

$$N = 6.81 \text{ standard drinks}$$

$\therefore$ Number of low alcohol bottles

$$= \frac{6.81}{0.8}$$

$$= 8.51$$

$\therefore$ Max complete bottles to stay under 0.05

$$= 8$$

17. Algebra, STD2 A1 2005 HSC 24b

$$\begin{aligned} \text{i. } D &= \frac{2A}{15} \\ &= \frac{2 \times 9}{15} \\ &= 1.2 \text{ mL} \end{aligned}$$

$\therefore$ George should be given a dosage of 1.2 mL

ii. Find A when $D = 4$ mL

$$4 = \frac{2 \times A}{15}$$

$$2A = 60$$

$$A = 30$$

$\therefore$ Sam is 30 months old and is 21 months older than George.

18. Algebra, STD2 A1 2018 HSC 28e

$$\begin{aligned} 70 \text{ km/hr} &= 70\,000 \text{ m/hr} \\ &= \frac{70\,000}{60 \times 60} \text{ m/sec} \\ &= 19.44... \text{ m/sec} \end{aligned}$$

$\therefore$ Total stopping distance

$$\begin{aligned} &= \text{reaction time distance} + \text{braking distance} \\ &= 1.5 \times 19.44... + 0.01 \times 70^2 \\ &= 78.166... \\ &= 78 \text{ m (nearest m)} \end{aligned}$$

♦ Mean mark 46%.

19. Algebra, STD2 A1 SM-Bank 4

$$\begin{aligned} 35 \text{ km/hr} &= 35\,000 \text{ m/hr} \\ &= \frac{35\,000}{60 \times 60} \text{ m/sec} \\ &= 9.722\ldots \text{ m/sec} \end{aligned}$$

$$\begin{aligned} \therefore \text{Total stopping distance} &= \text{Reaction time distance} + \text{braking distance} \\ &= 1.25 \times 9.722\ldots + 5.3 \\ &= 17.452\ldots \\ &= 17.5 \text{ metres (to 1 d.p.)} \end{aligned}$$

20. Algebra, STD2 A1 2015 HSC 30d

$$\begin{aligned} 110 \text{ km/hr} &= 110\,000 \text{ m/hr} \\ &= \frac{110\,000}{60 \times 60} \text{ m/sec} \\ &= 30.555\ldots \text{ m/sec} \end{aligned}$$

♦ Mean mark 34%.

$$\begin{aligned} \text{Reaction time distance} &= 2 \times 30.555\ldots \\ &= 61.11\ldots \text{ metres} \end{aligned}$$

$$\begin{aligned} \therefore \text{Stopping distance} &= \text{Reaction time distance} + \text{braking distance} \\ &= 61.11\ldots + 59.2 \\ &= 120.311\ldots \\ &= 120.3 \text{ metres (to 1 d.p.)} \end{aligned}$$